Python Basics: Print, Input and Variables

Your first Python program, explained step by step.

Python | Beginner | Ages 8–16

💡 The Big Idea

- Python is a popular, beginner friendly language used for games, websites, AI and science.
- A program is a list of instructions that Python follows from top to bottom.
- `print()` shows something on the screen. `input()` asks the user to type.
- A variable is a name that stores a value, like `age = 12`.
- Python cares about spelling, capital letters and indentation.

🔤 Data Types

- `int` is a whole number: 7, -3, 100.
- `float` is a number with a decimal point: 3.14, 0.5.
- `str` is text in quotes: "hello" or 'hello'.
- `bool` is True or False.
- `type(x)` tells you what kind of value `x` is.

➕ Operators

- `+ - * /` add, subtract, multiply, divide (divide always gives a float).
- `//` whole number division, `%` the remainder, `**` the power.
- `+` joins two strings: "Hi " + "Mia".
- Use brackets to control the order: (2 + 3) * 4.

📝 Rules and Habits

- Variable names use letters, numbers and underscores, and cannot start with a number.
- Use clear names: `player_score`, not `x`.
- Python is case sensitive: `Score` and `score` are different.
- Comments start with `#`. Python ignores them.
- `input()` always gives text. Use `int()` or `float()` to turn it into a number.

💻 Code Examples

Hello, world! python

```python
print("Hello, world!")
print("I am learning Python")
```

▶ **Output:**

```
Hello, world!
I am learning Python
```

Each print goes on a new line.

Variables and types python

```python
name = "Mia"
age = 12
height = 1.45
is_student = True
print(name, age, height, is_student)
print(type(age), type(height), type(name), type(is_student))
```

▶ **Output:**

```
Mia 12 1.45 True
<class 'int'> <class 'float'> <class 'str'> <class 'bool'>
```

type() shows the kind of value.

Maths with operators python

```python
a = 17
b = 5
print(a + b, a - b, a * b)
print(a / b)
print(a // b, a % b)
print(2 ** 10)
```

▶ **Output:**

```
22 12 85
3.4
3 2
1024
```

17 divided by 5 is 3 remainder 2.

Asking the user (input) python

```python
name = input("What is your name? ")
age = int(input("How old are you? "))
print("Hello", name)
print("Next year you will be", age + 1)
```

▶ **Output:**

```
What is your name? Mia
How old are you? 12
Hello Mia
Next year you will be 13
```

int() turns the typed text into a number so we can add 1.

f-strings (the easy way to mix text and values) python

```python
name = "Mia"
score = 95
print(f"{name} scored {score} points")
print(f"Double score: {score * 2}")
```

▶ **Output:**

```
Mia scored 95 points
Double score: 190
```

Put f before the quotes, and write values inside curly brackets.

TypeError: text plus number python

```python
age = 12
print("I am " + age)
```

▶ **Error message:**

```
TypeError: can only concatenate str (not "int") to str
```

Fix: print("I am " + str(age)) or use an f-string.

📚 Key Terms

Program
A list of instructions for the computer.

Integer
A whole number.

Variable
A name that stores a value.

Float
A number with a decimal point.

String
Text in quotes.

Boolean
True or False.

Operator
A symbol that does a calculation.

Output
What the program shows.

Input
Information the user types in.

Comment
A note for people, starting with #.

🧮 Quick Reference

```
print("Hi")
name = "Mia"
age = int(input("Age? "))
print(f"{name} is {age}")
```

```
10 // 3  # 3
10 % 3   # 1
2 ** 3   # 8
# this is a comment
```

✏️ Worked Examples

Q: Ask for a name and say hello.
A: name = input("Name? ") stores the answer. print("Hello", name) shows it.

Q: Add two numbers typed by the user.
A: Use a = int(input()) and b = int(input()), then print(a + b). int() is needed because input() gives text.

Q: Work out the area of a rectangle with width 6 and height 4.
A: width = 6, height = 4, area = width * height, print(area) shows 24.

🚫 Common Mistakes

- Forgetting quotes around text: `print(Hello)` causes a NameError.
- Forgetting brackets: `print "Hi"` is old Python 2 style.
- Adding text and numbers without `str()` or an f-string.
- Using `=` for comparing. `=` stores a value and `==` compares.
- Mixing up capital letters, like `Print` instead of `print`.

👨‍👩‍👧 Parent Corner

- You can try Python free online with a browser based editor. No install is needed.
- Ask your child to make a program that asks your name and greets you.
- Ask: What type is this value? Number, text or True/False?

- Encourage typing code themselves rather than copying, and reading each error message together.
- Celebrate every working program, however small.

🎯 Quick Check

1. What does print("Hi") do?

✅ It shows Hi on the screen.

2. What type does input() always give?

✅ A string (text).

3. What is 17 % 5?

✅ 2, the remainder.

4. How do you turn typed text into a whole number?

✅ int(input()).

5. What starts a comment?

✅ The # symbol.